Reynolds Number and Flow Regimes

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Fluid Mechanics Lecture Notes

Learning Objectives

By the end of this lecture, students should be able to:

- explain Reynolds number as a comparison of inertial and viscous effects;
- distinguish between dynamic viscosity and kinematic viscosity;
- select the correct representative velocity and characteristic length;
- verify that Reynolds number is dimensionless;
- classify circular-pipe flow as laminar, transitional, or turbulent;
- calculate Reynolds number and a limiting laminar flow rate from volume-flow data.

1 Why Reynolds Number Matters

Reynolds number is a dimensionless quantity that compares two competing effects in fluid flow:

$$\boxed{\text{inertial effects} \quad \text{compared with} \quad \text{viscous effects}}$$

Inertia tends to carry fluid motion forward and preserve disturbances. Viscosity resists relative motion between neighboring fluid layers and tends to smooth disturbances.

Reynolds number is used to:

- identify whether a flow is likely to be laminar, transitional, or turbulent;
- determine which friction-factor relation or Moody-chart region applies;
- compare dynamically similar flows of different sizes and speeds;
- support heat-transfer correlations involving Prandtl and Nusselt numbers;
- estimate where transition may begin in internal and external flows.

2 Definition and Physical Meaning

The Reynolds number can be written using dynamic viscosity:

$$\boxed{\mathrm{Re} = \frac{\rho U L}{\mu}}$$

Since kinematic viscosity is

$$\nu = \frac{\mu}{\rho},$$

the same quantity can also be written as

$$\boxed{\mathrm{Re} = \frac{U L}{\nu}}$$

The physical origin of this ratio can be seen by comparing characteristic stress scales. Inertial stress scales as

$$\tau_{\text{inertia}} \sim \rho U^2,$$

while viscous stress scales as

$$\tau_{\text{viscous}} \sim \mu \frac{U}{L}.$$

Their ratio is

$$\frac{\tau_{\text{inertia}}}{\tau_{\text{viscous}}} \sim \frac{\rho U^2}{\mu U/L} = \frac{\rho U L}{\mu} = \mathrm{Re}.$$

Physical Interpretation

- **Low Reynolds number:** viscous effects are relatively strong and disturbances are readily damped.
- **High Reynolds number:** inertial effects are relatively strong and disturbances can persist or grow.
- Reynolds number has no units, but its interpretation depends on the geometry and on how U and L are defined.

3 Variables in Reynolds Number

3.1 Fluid Properties

The density ρ, dynamic viscosity μ, and kinematic viscosity ν are fluid properties. In introductory calculations, they are often treated as constant.

Viscosity can change strongly with temperature, especially for oils. The properties must therefore be evaluated at the stated operating temperature.

Symbol	Meaning and SI unit
ρ	density, kg/m^3
μ	dynamic viscosity, Pas
$\nu = \mu/\rho$	kinematic viscosity, m^2/s
U	representative velocity, m/s
L	characteristic length, m

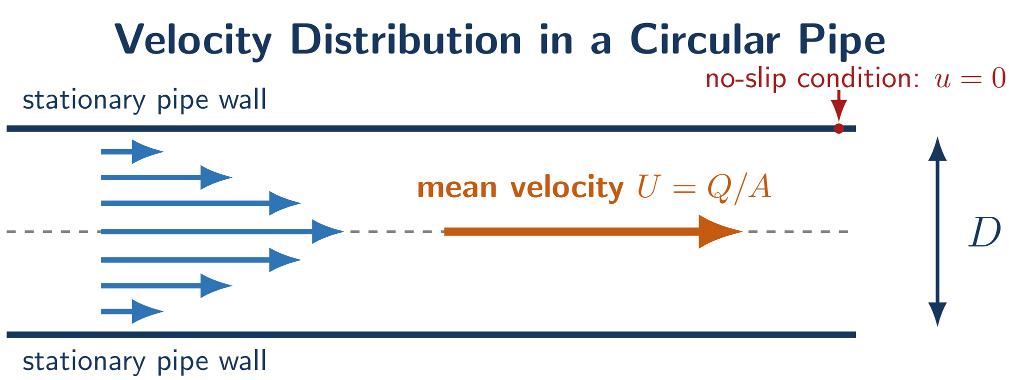
3.2 Representative Velocity

Velocity is generally not uniform across a pipe. Because of the no-slip condition, the fluid velocity is zero at the wall and larger away from the wall.

For internal flow, the velocity in Reynolds number normally refers to the **cross-sectional mean velocity**:

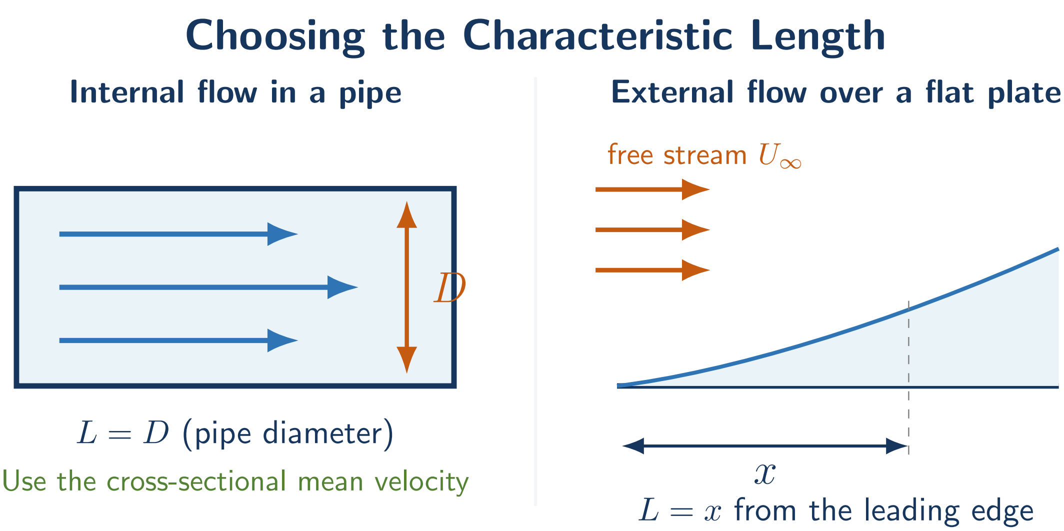
$$\boxed{U = \frac{Q}{A}}$$

where Q is the volume flow rate and A is the cross-sectional area.



3.3 Characteristic Length

The characteristic length depends on the type of flow.



Flow situation	Representative velocity and characteristic length
Circular pipe	mean velocity $U = Q/A$, pipe diameter $L = D$
Noncircular duct	mean velocity $U = Q/A$, hydraulic diameter $L = D_h = 4A/P_w$
Flat plate	free-stream velocity U_∞, distance $L = x$ from the leading edge
Flow around an object	free-stream velocity U_∞, a specified object dimension such as diameter

Here P_w is the wetted perimeter. The chosen velocity and length must always be stated with the Reynolds number.

4 Dimensional Analysis

Using SI base units,

$$[\rho] = \frac{\text{kg}}{\text{m}^3}, \qquad [U] = \frac{\text{m}}{\text{s}}, \qquad [L] = \text{m}, \qquad [\mu] = \frac{\text{kg}}{\text{ms}}.$$

Substitution into the definition gives

$$[\text{Re}] = \frac{(\text{kg}/\text{m}^3)(\text{m}/\text{s})(\text{m})}{\text{kg}/\text{ms}} = 1.$$

Therefore,

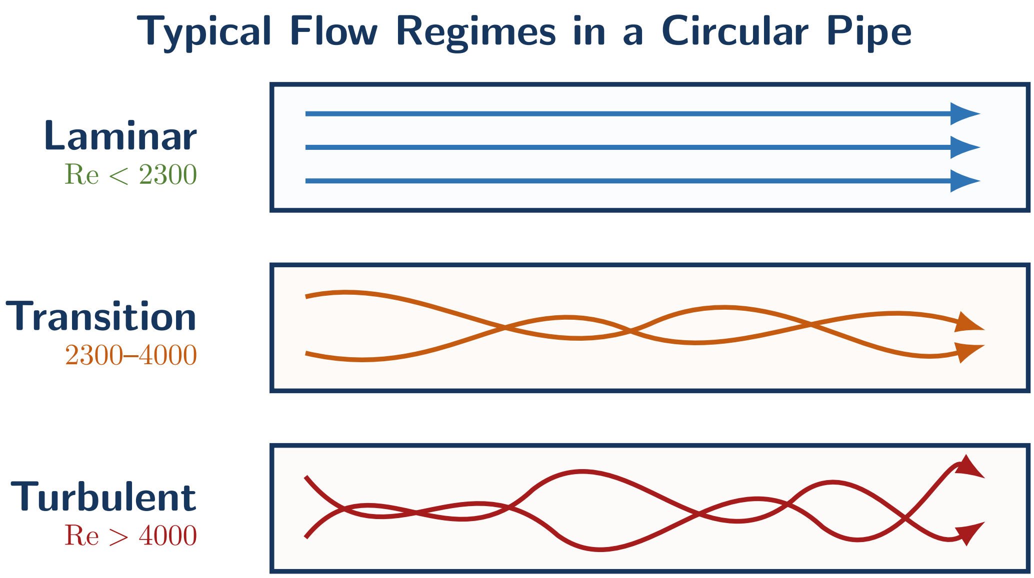
$$\boxed{\text{Re is dimensionless.}}$$

The same cancellation occurs when using any consistent unit system.

5 Flow Regimes

For steady flow in a smooth circular pipe, the commonly used guidelines are:

Reynolds number	Regime	Physical description
$\mathrm{Re}_D < 2300$	Laminar	orderly layers and weak cross-stream mixing
$2300 \leq \mathrm{Re}_D \leq 4000$	Transitional	highly sensitive to disturbances
$\mathrm{Re}_D > 4000$	Turbulent	strong fluctuations and mixing



A turbulent flow still has an overall mean direction even though the instantaneous particle motion contains irregular fluctuations.

For external flow over a smooth flat plate, a commonly used transition estimate is

$$\mathrm{Re}_{x,\mathrm{cr}} \approx 5 \times 10^5,$$

where

$$\mathrm{Re}_x = \frac{\rho U_\infty x}{\mu} = \frac{U_\infty x}{\nu}.$$

Critical Reynolds Numbers Depend on the Flow

The values 2300 and 4000 are guidelines for circular-pipe flow. They should not be applied automatically to a flat plate, an open channel, a jet, or flow around an object. Transition also depends on inlet disturbances, surface roughness, vibration, and free-stream turbulence.

6 Volume Flow Rate and Mean Velocity

Volume flow rate can be measured by collecting a known volume over a measured time:

$$Q = \frac{\Delta \mathcal{V}}{\Delta t}.$$

For example, if a 1 L bottle fills in 100 s, then

$$Q = \frac{1\,\mathrm{L}}{100\,\mathrm{s}} = 0.01\,\mathrm{L/s} = 1.0 \times 10^{-5}\,\mathrm{m^3/s}.$$

Since volume equals area times distance, volume per time equals area times velocity:

$$\boxed{Q = UA}$$

For a circular pipe,

$$A = \frac{\pi D^2}{4}, \qquad U = \frac{Q}{A} = \frac{4Q}{\pi D^2}.$$

Substituting into the pipe Reynolds number gives

$$\boxed{\mathrm{Re}_D = \frac{\rho U D}{\mu} = \frac{4\rho Q}{\pi \mu D} = \frac{4Q}{\pi \nu D}}$$

This form is especially useful when the pipe diameter and volume flow rate are known.

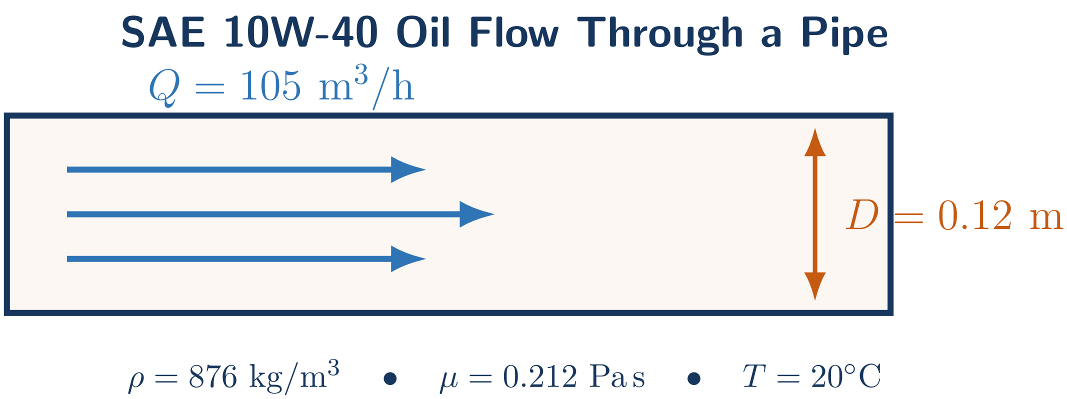
7 Worked Example 1: SAE 10W-40 Oil

Problem

SAE 10W-40 oil at 20°C flows through a 0.12 m-diameter pipe at a volume flow rate of $105\,\text{m}^3/\text{h}$. Use

$$\rho = 876\,\text{kg/m}^3, \qquad \mu = 0.212\,\text{Pa s}.$$

Determine the flow regime. Then find the maximum flow rate that satisfies the conservative laminar-flow limit $\text{Re}_D = 2000$.



First convert the volume flow rate:

$$Q = 105\,\text{m}^3/\text{h}\left(\frac{1\,\text{h}}{3600\,\text{s}}\right) = 0.02917\,\text{m}^3/\text{s}.$$

The cross-sectional area and mean velocity are

$$A = \frac{\pi D^2}{4} = \frac{\pi(0.12\,\text{m})^2}{4} = 0.01131\,\text{m}^2,$$

$$U = \frac{Q}{A} = \frac{0.02917\,\text{m}^3/\text{s}}{0.01131\,\text{m}^2} = 2.579\,\text{m/s}.$$

Now calculate Reynolds number:

$$\text{Re}_D = \frac{\rho U D}{\mu}$$

$$\text{Re}_D = \frac{(876\,\text{kg/m}^3)(2.579\,\text{m/s})(0.12\,\text{m})}{0.212\,\text{Pa s}} = 1279.$$

Therefore,

$$\boxed{\mathrm{Re}_D \approx 1279 < 2300 \quad \Longrightarrow \quad \text{laminar flow}}$$

For the conservative limiting flow rate, set

$$\mathrm{Re}_{D,\max} = 2000.$$

Solve for the maximum mean velocity:

$$U_{\max} = \frac{\mathrm{Re}_{D,\max}\mu}{\rho D} = \frac{(2000)(0.212\,\mathrm{Pa\,s})}{(876\,\mathrm{kg/m^3})(0.12\,\mathrm{m})} = 4.034\,\mathrm{m/s}.$$

Then

$$Q_{\max} = U_{\max} A$$

$$Q_{\max} = (4.034\,\mathrm{m/s})(0.01131\,\mathrm{m^2}) = 0.04562\,\mathrm{m^3/s}.$$

Converting to cubic meters per hour,

$$\boxed{Q_{\max} = 164.2\,\mathrm{m^3/h}}$$

Important Observation

For a fixed fluid and pipe diameter, Reynolds number is directly proportional to volume flow rate. Therefore, the limiting flow rate can also be checked using

$$\frac{Q_{\max}}{Q} = \frac{\mathrm{Re}_{\max}}{\mathrm{Re}}.$$

Oil properties vary by product and temperature, so a different property table will produce a different numerical result.

8 Worked Example 2: Water Flow in a Small Pipe

Problem

Water at 20°C flows through a 25 mm-diameter pipe at $Q = 2.0 \times 10^{-4}\,\mathrm{m^3/s}$. Use $\nu = 1.00 \times 10^{-6}\,\mathrm{m^2/s}$. Determine the Reynolds number and flow regime.

The pipe area and mean velocity are

$$A = \frac{\pi D^2}{4} = \frac{\pi(0.025\,\mathrm{m})^2}{4} = 4.909 \times 10^{-4}\,\mathrm{m}^2,$$

$$U = \frac{Q}{A} = \frac{2.0 \times 10^{-4}\,\mathrm{m}^3/\mathrm{s}}{4.909 \times 10^{-4}\,\mathrm{m}^2} = 0.407\,\mathrm{m/s}.$$

Using the kinematic-viscosity form,

$$\mathrm{Re}_D = \frac{UD}{\nu} = \frac{(0.407\,\mathrm{m/s})(0.025\,\mathrm{m})}{1.00 \times 10^{-6}\,\mathrm{m}^2/\mathrm{s}} = 1.02 \times 10^4.$$

$$\boxed{\mathrm{Re}_D \approx 10{,}200 > 4000 \quad \Longrightarrow \quad \text{turbulent flow}}$$

9 Worked Example 3: Transition Over a Flat Plate

Problem

Air flows over a smooth flat plate at $U_\infty = 12\,\mathrm{m/s}$. Use $\nu = 1.50 \times 10^{-5}\,\mathrm{m}^2/\mathrm{s}$ and estimate the distance from the leading edge at which $\mathrm{Re}_x = 5.0 \times 10^5$.

For a flat plate,

$$\mathrm{Re}_x = \frac{U_\infty x}{\nu}.$$

Solving for x,

$$x = \frac{\mathrm{Re}_x \nu}{U_\infty}$$

$$x = \frac{(5.0 \times 10^5)(1.50 \times 10^{-5}\,\mathrm{m}^2/\mathrm{s})}{12\,\mathrm{m/s}} = 0.625\,\mathrm{m}.$$

$$\boxed{x_{\mathrm{cr}} \approx 0.63\,\mathrm{m}}$$

This is an estimate. Surface roughness and free-stream disturbances can move the actual transition location upstream or downstream.

10 A Reliable Problem-Solving Procedure

1. Identify the flow geometry: pipe, duct, flat plate, jet, or external body.

2. Select the representative velocity and characteristic length.
3. Obtain ρ and μ, or obtain ν, at the stated temperature.
4. Convert all quantities to one consistent unit system.
5. If volume flow rate is given, calculate $U = Q/A$.
6. Write the Reynolds-number equation before substituting values.
7. Calculate Re and state the geometry-specific flow-regime guideline.
8. Check whether the result is physically reasonable.

Common Choices

Situation	Reynolds-number definition
Circular pipe	$\mathrm{Re}_D = \rho U D/\mu = UD/\nu$
Noncircular duct	$\mathrm{Re}_{D_h} = \rho U D_h/\mu$, where $D_h = 4A/P_w$
Flat plate at location x	$\mathrm{Re}_x = \rho U_\infty x/\mu$
Known circular-pipe flow rate	$\mathrm{Re}_D = 4\rho Q/(\pi \mu D)$
Conservative laminar pipe limit	use $\mathrm{Re}_D = 2000$ when specified

11 Common Mistakes

1. **Using a local pipe velocity instead of the mean velocity.** For internal flow, use $U = Q/A$ unless another velocity is explicitly specified.
2. **Choosing the wrong characteristic length.** Use D for a circular pipe, D_h for a noncircular duct, and x for local flow over a flat plate.
3. **Mixing viscosity forms.** Use either $\rho UL/\mu$ or UL/ν; do not insert ν into the first form.
4. **Using fluid properties at the wrong temperature.**
5. **Forgetting to convert hours to seconds or millimeters to meters.**
6. **Applying pipe thresholds to every geometry.**
7. **Confusing fully developed flow with turbulent flow.** Fully developed describes whether the velocity-profile shape changes along the pipe; it does not identify the regime.

12 In-Class Concept Questions

Check Your Understanding

1. If the volume flow rate doubles in the same pipe with the same fluid, what happens to Re_D?
2. If kinematic viscosity increases while U and L remain constant, what happens to Reynolds number?
3. Why is the centerline velocity not normally used for the pipe Reynolds number?
4. For a flat plate, why does Re_x increase with distance from the leading edge?
5. Can a fully developed pipe flow be laminar? Explain.

13 Practice Problems

Problem 1: Water in a Small Pipe

Water at 20°C flows through a 20 mm-diameter pipe at $Q = 3.0 \times 10^{-5}\,\mathrm{m^3/s}$. Use $\nu = 1.00 \times 10^{-6}\,\mathrm{m^2/s}$. Determine the mean velocity, Reynolds number, and flow regime.

Problem 2: Maximum Laminar Oil Flow

Oil with $\nu = 2.50 \times 10^{-4}\,\mathrm{m^2/s}$ flows through a 50 mm-diameter pipe. Using $\mathrm{Re}_{D,\max} = 2000$, determine the maximum volume flow rate for conservative laminar operation.

Problem 3: Flow-Rate Scaling

The flow rate in a fixed pipe carrying the same fluid increases by 50%. By what percentage does the pipe Reynolds number change?

Problem 4: Flat-Plate Transition

Air flows over a smooth flat plate at $U_\infty = 15\,\mathrm{m/s}$. Use $\nu = 1.50 \times 10^{-5}\,\mathrm{m^2/s}$ and $\mathrm{Re}_{x,\mathrm{cr}} = 5.0 \times 10^5$. Estimate the transition distance from the leading edge.

Problem 5: Flow Measurement

A 2.0 L container fills in 40 s from an 8 mm-diameter water line. Use $\nu = 1.00 \times 10^{-6}\,\mathrm{m^2/s}$. Determine Q, the mean velocity, and the Reynolds number.

14 Answers to Practice Problems

1.

$$A = \frac{\pi D^2}{4} = 3.142 \times 10^{-4}\,\mathrm{m^2}, \qquad U = \frac{Q}{A} = 0.0955\,\mathrm{m/s}.$$

$$\mathrm{Re}_D = \frac{UD}{\nu} = \frac{(0.0955)(0.020)}{1.00 \times 10^{-6}} = \boxed{1910,\ \text{laminar}}.$$

2.
$$Q_{\max} = \frac{\pi \nu D \mathrm{Re}_{D,\max}}{4}$$
$$Q_{\max} = \frac{\pi(2.50 \times 10^{-4})(0.050)(2000)}{4} = \boxed{1.96 \times 10^{-2}\,\mathrm{m^3/s}}.$$

3. Since the fluid and diameter are unchanged,
$$\mathrm{Re}_D \propto Q.$$
Therefore, the Reynolds number also $\boxed{\text{increases by 50\%}}$.

4.
$$x_{\mathrm{cr}} = \frac{\mathrm{Re}_{x,\mathrm{cr}}\nu}{U_\infty} = \frac{(5.0 \times 10^5)(1.50 \times 10^{-5})}{15} = \boxed{0.50\,\mathrm{m}}.$$

5.
$$Q = \frac{2.0 \times 10^{-3}\,\mathrm{m^3}}{40\,\mathrm{s}} = 5.0 \times 10^{-5}\,\mathrm{m^3/s}.$$
$$A = \frac{\pi(0.008)^2}{4} = 5.027 \times 10^{-5}\,\mathrm{m^2}, \qquad U = \frac{Q}{A} = 0.995\,\mathrm{m/s}.$$
$$\mathrm{Re}_D = \frac{UD}{\nu} = \frac{(0.995)(0.008)}{1.00 \times 10^{-6}} = \boxed{7960,\ \text{turbulent}}.$$

15 Summary

Key Takeaways

- Reynolds number compares inertial effects with viscous effects:
$$\mathrm{Re} = \frac{\rho U L}{\mu} = \frac{UL}{\nu}.$$
- For internal flow, use the cross-sectional mean velocity $U = Q/A$.
- For a circular pipe, the characteristic length is the diameter D:
$$\mathrm{Re}_D = \frac{\rho U D}{\mu} = \frac{4\rho Q}{\pi \mu D}.$$
- Typical circular-pipe guidelines are laminar below 2300, transitional from 2300 to 4000, and turbulent above 4000.
- Critical values depend on geometry and flow conditions.
- Always state the representative velocity, characteristic length, fluid properties, and geometry with the Reynolds number.